A Real-time Multimodal Adaptive Mixed Reality System for Procedural Task Guidance

Anonymous Author(s)

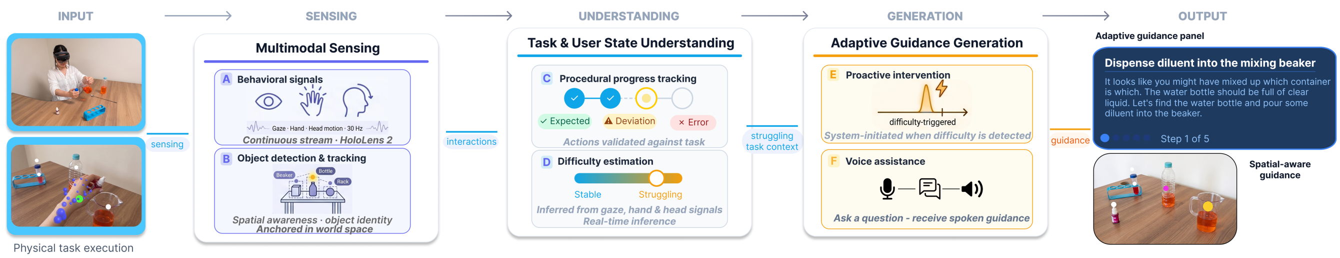

Figure 1: Overview of the proposed adaptive mixed reality guidance system. Multimodal sensing captures behavioral signals including gaze, hand, and head motion (A) and tracks physical objects in world space (B). Task and user state understanding continuously monitors procedural progress and classifies action outcomes (C) while estimating difficulty from interaction dynamics (D). Adaptive guidance generation triggers proactive intervention when difficulty is detected (E) and responds to user voice queries (F), delivering context-aware assistance through a spatially grounded MR guidance panel.

Abstract

Procedural tasks often require users to interpret instructions, monitor task progress, and recover from errors while interacting with physical objects. Mixed reality (MR) can support such tasks by presenting guidance directly within the user's workspace. However, providing effective assistance remains challenging as user behavior, task progress, and support requirements may vary throughout task execution. We present a real-time multimodal adaptive MR system for procedural task guidance that integrates semantic task tracking, user-state estimation, and adaptive assistance within a unified closed-loop architecture. The system represents procedures as dependency-aware task graphs, continuously tracks task progress through multimodal sensing of user interactions, and estimates procedural difficulty from gaze, hand, and head-motion signals. These signals are used to dynamically adapt guidance content, timing, and presentation through an LLM-based assistance module. We implemented the system on Microsoft HoloLens 2 and evaluated it through technical validation and a user study involving 20 participants performing procedural tasks of varying complexity. Results demonstrate robust closed-loop operation, achieving an action-recognition F1 score of 0.97 and 100% step-completion accuracy while maintaining real-time task-state monitoring and feedback delivery. User study results indicate that adaptive guidance reduced procedural difficulties and recovery time following task deviations, particularly during more complex procedures. Our findings demonstrate the effectiveness of combining task-state and user-state awareness to support personalized procedural guidance in mixed reality.

CCS Concepts

• **Human-centered computing → Mixed / augmented reality.**

Keywords

Mixed Reality, Procedural Task Guidance, Adaptive Assistance, User-State Estimation, Task-State Tracking, Multimodal Interaction, Large Language Models

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1 Introduction

Procedural tasks require users to interpret instructions, perform ordered physical actions, monitor their progress, and recover from errors while interacting with real objects. Such tasks are common in laboratory work, technical maintenance, assembly, medical assistance, and everyday skill acquisition. In these settings, mistakes can reduce efficiency, compromise safety, or prevent successful task completion. Conventional instructional materials such as paper manuals, videos, and static checklists provide useful reference information, but they do not actively interpret what the user is doing or adapt support when the user deviates from the intended procedure.

Mixed reality (MR) offers a promising platform for procedural guidance because it can spatially align digital instructions with the physical workspace while preserving embodied interaction with real tools and objects [1, 11, 16]. Head-mounted MR devices can also capture multimodal behavioral signals, including gaze,

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hand motion, head pose, object detections, and interaction events [6, 7, 21]. These capabilities make MR well suited for situated, step-by-step procedural assistance. However, moving from static MR instructions to adaptive procedural guidance remains a challenging systems problem. An effective system must not only display the next instruction, but also reason about the current procedural state, interpret whether observed actions are expected or problematic, estimate when the user is experiencing difficulty, and provide assistance without unnecessarily interrupting successful task execution.

Prior MR and XR guidance systems have demonstrated the value of spatial cues and interactive instructions, but many systems remain limited to fixed step sequences or interface-level adaptation [1, 10, 16]. Similarly, egocentric action-recognition and interactive-assistance methods can identify observed activities, but procedural assistance requires interpreting those observations with respect to task dependencies, completion conditions, and possible recovery paths [9, 12, 20]. User-state adaptation introduces an additional challenge: latent constructs such as cognitive load, confusion, or affective state are difficult to infer robustly in real-world MR environments [13, 18]. For procedural guidance, we instead focus on observable procedural difficulty, estimated from multimodal behavioral signals that can be sensed during task execution.

To address these challenges, we propose a real-time task- and user-state-aware adaptive MR system for procedural task guidance. The system represents procedures as dependency-aware task graphs generated from natural-language protocols, tracks user progress through semantic interpretation of multimodal interaction signals, estimates procedural difficulty from gaze, hand, and head-motion behavior, and adapts guidance through an LLM-based assistance module. The design is informed by recent work on adaptive multimodal assistants and LLM-grounded XR systems, while focusing on end-to-end procedural task guidance in physical MR settings [2, 17, 19, 25]. The system is implemented on Microsoft HoloLens 2 and is designed as a closed-loop runtime architecture: it observes user behavior, updates task state, estimates user state, and delivers contextual assistance in the user's workspace.

This paper makes the following contributions:

- A closed-loop MR architecture that integrates semantic procedural task tracking, behavioral user-state estimation, and adaptive guidance generation for physical procedural tasks.
- A real-time semantic task-tracking framework that infers procedural state from multimodal interactions with physical objects, enabling monitoring of task progress and procedural deviations during task execution.
- A multimodal procedural difficulty estimation approach based on gaze, hand, and head-motion signals available from a head-mounted MR device.
- An LLM-based adaptive assistance module that generates context-aware guidance conditioned on procedural task state, estimated user difficulty, and user queries.
- A system performance evaluation and user study demonstrating reliable real-time operation and evidence that adaptive guidance is particularly beneficial during more complex procedural tasks.

2 Related Work

2.1 Procedural Guidance in Mixed Reality

Mixed reality is well suited to procedural guidance because it can spatially register digital instructions with the physical workspace while preserving direct manipulation of real tools [11]. Existing systems have applied this capability to assembly [16], skills training [1], and immersive education [10], demonstrating the value of situated spatial cues over paper- or video-based references. However, these systems largely deliver predefined step sequences and adapt, if at all, only at the interface level; they do not reason about whether an observed action is expected, premature, or recoverable within the procedure. Broader surveys of XR and AI similarly note that adaptive, behavior-aware assistance remains an open direction rather than a solved capability [6, 7, 19].

2.2 Egocentric Sensing, Action Recognition, and Task Tracking

Head-mounted MR devices expose rich egocentric signals—gaze, hand pose, head motion, and object detections—that can be used to interpret user activity. Large-scale egocentric datasets such as HoloAssist [9, 20] have advanced action understanding for real-world procedural interaction, and recent work has explored multimodal fine-grained training assistants in MR [12]. Yet recognizing *what* action occurred is not equivalent to interpreting what it *means* relative to task dependencies, completion conditions, and recovery paths. Procedural assistance therefore requires a semantic layer that maps recognized actions onto an explicit task representation, motivating the dependency-aware task-graph tracking used in our system.

2.3 User-State Estimation from Behavioral Signals

Adaptive XR systems increasingly attempt to personalize support by estimating user state, but latent constructs such as cognitive load or affect are difficult to infer robustly outside controlled conditions, and approaches relying on physiological sensing transfer poorly to unconstrained MR procedural tasks [13]. Most relevant to our work, Stiber et al. [18] detect confusion during physical tasks from behavioral signals, framing confusion as an observable "stuck" state rather than a direct cognitive measurement. Our approach shares this behaviorally grounded framing but differs in three respects central to a deployable guidance system: it performs real-time *streaming* inference over raw multimodal sequences rather than offline classification of aggregated hand-crafted features; the resulting difficulty estimate is embedded in a *closed loop* that drives runtime guidance adaptation; and it is evaluated within a complete system through a *user study*, rather than as a standalone detection task.

2.4 LLM-Based and Proactive Procedural Assistance

Large language models are increasingly integrated into XR to support contextual reasoning and natural-language guidance [19]. Platform-oriented efforts such as XaiR [17] and SIGMA [2] connect LLM reasoning and multimodal sensing to the physical world, while AMMA [25] performs automated state tracking with user-model-directed guidance planning. A recent wave of systems targets proactive and procedural assistance specifically: Satori [8] models user intention through a belief–desire–intention formulation; MRPilot [24] focuses on responsive navigation of general procedural tasks; XR-Objects [5] emphasizes object-aware interaction and grounding; Pro^{2}Assist [22] provides continuous step-aware proactive assistance from egocentric perception; ProMemAssist [14] times interventions through working-memory modeling; and ProAgent [23] coordinates on-demand sensory context for proactive LLM agents. These systems mark important progress, but each tends to emphasize a single primary axis of adaptation—intention inference, procedural navigation, object grounding, intervention timing, or multimodal reasoning.

In contrast, the proposed system integrates semantic procedural task tracking with behaviorally grounded procedural-difficulty estimation within a single real-time closed loop on commodity MR hardware. Guidance is conditioned not only on the current procedural step, but also on whether the user appears to be experiencing elevated difficulty during execution. We position the contribution not as a generic XR guidance interface, but as a *difficulty-aware adaptive procedural assistance framework*, evaluated end-to-end over real physical tasks through a controlled user study.

3 System Overview

3.1 Design Goals and Considerations

We designed the system as a closed-loop runtime that observes user behavior, interprets its procedural meaning, estimates difficulty, and adapts guidance during execution. From the limitations identified above, we derived five design goals:

- **G1. Real-time semantic procedural tracking over physical objects.** Interactions must be interpreted relative to task dependencies and recovery paths, not labeled as isolated actions.
- **G2. Non-invasive multimodal sensing.** The system should rely only on gaze, hand, head, and object signals available on commodity MR hardware, without intrusive physiological instrumentation.
- **G3. Behaviorally grounded user-state estimation.** Adaptation should be driven by *observable procedural difficulty* inferred from interaction dynamics, rather than by latent cognitive or affective state.
- **G4. Reactive and proactive adaptive guidance.** Guidance must serve both user-initiated queries and system-initiated proactive intervention, conditioning LLM output on runtime procedural context and estimated difficulty to adapt assistance content, timing, and presentation, improving contextual grounding of generated guidance and intervening only when warranted rather than during successful task execution.
- **G5. Complementary explicit and implicit adaptation signals.** Because procedural difficulty estimation remains inherently uncertain, adaptation should not rely on a single prediction source. Instead, explicit procedural context and implicit difficulty estimates are maintained as complementary inputs, allowing guidance decisions to remain interpretable and degrade gracefully when uncertainty increases.

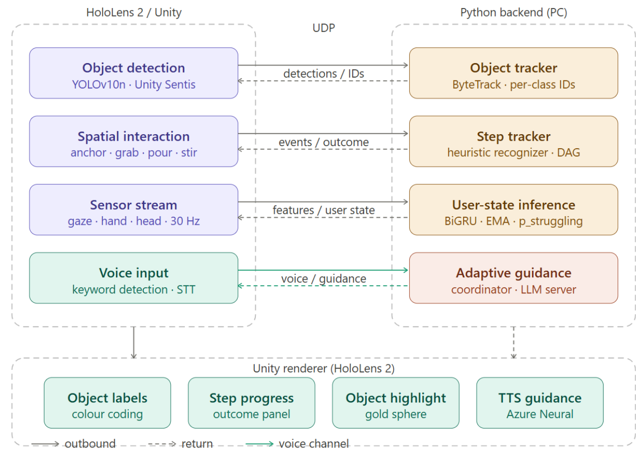

Figure 2: Closed-loop system pipeline for adaptive mixed reality procedural guidance.

To support real-time operation on resource-constrained MR hardware, we separate lightweight interaction processing from computationally heavier temporal and language-model inference. This distributed design balances responsiveness and deployment feasibility while preserving the modular separation between explicit and implicit adaptation signals. Consistent with the real-time requirements of the system, interaction processing relies on lightweight interpretable recognition mechanisms rather than computationally intensive end-to-end video understanding or vision–language models.

3.2 System Architecture

The system comprises four tightly coupled components that together form a closed loop (Fig. 2):

(1) **Procedural task modeling.** A natural-language protocol is converted by an LLM into a dependency-aware task graph (DAG) capturing steps, required objects, and allowable transitions (Sec. 4.1).
(2) **Semantic task-state tracking.** Multimodal interaction signals—object detections, hand–object cues, gaze, and MR events—are interpreted as semantic actions and validated against the graph, producing procedural outcomes (Sec. 4.2).
(3) **Behavioral user-state estimation.** Temporal gaze, hand, and head-motion sequences are processed by a recurrent model to estimate elevated procedural difficulty relative to stable interaction (Sec. 4.3).
(4) **Adaptive guidance.** A user voice query or a proactive trigger, together with the current procedural state, recent outcomes, interaction history, and estimated difficulty, conditions an LLM that returns verbal and visual guidance delivered through the MR interface (Sec. 4.4–4.5).

The framework is implemented as a distributed MR system consisting of an on-device sensing and interaction layer and backend services responsible for task tracking, user-state estimation, and

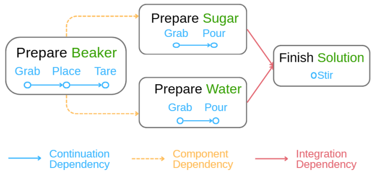

Figure 3: Dependency-aware procedural task graph showing continuation, component, and integration dependencies.

adaptive guidance generation. This design supports real-time operation while allowing computationally intensive inference to be performed outside the headset.

4 Adaptive MR Guidance System

4.1 LLM-Based Procedural Task Modeling

The system represents each procedure as a dependency-aware directed acyclic graph (DAG) generated from a natural-language protocol using a large language model during an offline initialization stage. Each node corresponds to a semantic procedural step and stores task-relevant metadata, including associated objects, dependency constraints, optional substeps, and expected procedural actions. Directed edges encode relationships between steps and define the set of valid transitions that may occur during task execution.

We adopt a graph-based representation rather than a linear checklist because procedural tasks often contain dependencies, parallel preparation activities, and recovery paths that cannot be adequately represented through a strictly sequential model. By explicitly modeling these relationships, the graph provides a structured procedural representation that supports reasoning about task progression, deviations, repeated actions, and recovery during execution.

To support procedural reasoning beyond simple step ordering, the graph distinguishes three dependency types (Fig. 3). *Continuation dependencies* capture sequential progression within a procedural branch. *Component dependencies* represent prerequisite preparation steps that must be completed before a dependent action can be performed. *Integration dependencies* model situations in which multiple procedural branches converge into a shared downstream step.

The resulting graph serves as the semantic foundation for the remainder of the framework. During runtime, observed user interactions are continuously interpreted relative to the graph to determine procedural progress and validate whether actions are expected, premature, repeated, incorrect, or part of a recovery sequence. The graph also provides the procedural context supplied to the adaptive guidance module, including the current task state, recent procedural outcomes, and available future transitions.

By automatically generating the procedural model from textual protocols, the framework can be rapidly adapted to new procedures while preserving the same task-tracking and guidance mechanisms without requiring manually authored task-specific logic.

4.2 Semantic Task-State Tracking

At runtime, the system interprets multimodal interaction signals as semantic procedural actions and validates them against the current task-graph state. The tracking pipeline spans the on-device frontend, which performs object detection, hand–object interaction sensing, and spatial anchoring, and a backend tracker that performs semantic action recognition, dependency-aware validation, and outcome classification.

4.2.1 *Object Detection and Tracking.*

Object detection runs on-device using YOLOv10n executed through Unity Sentis, building on the open-source HoloLens object-detection pipeline of Dangberg et al. [4], which we adapted and extended for procedural interaction. Because several laboratory objects were poorly represented in the original COCO distribution, we fine-tuned task-specific detectors on images captured directly from the workspace with the HoloLens 2 (mAP@50 = 0.99 on the validation set). Two-dimensional detections are projected into the workspace using sphere casts from the camera origin through each detection center, yielding world-space estimates for indicators, labels, and interaction colliders.

Raw frame-by-frame detections exhibit positional jitter and intermittent identity switching. To stabilize identity, detections are forwarded over UDP to a backend service running ByteTrack [26] in per-class mode, which associates detections across frames using Kalman filtering and IoU matching and returns persistent `(track_id, class)` identities; when no valid association is available, the system falls back to class-based nearest-neighbor matching, and stale tracks are reset after prolonged inactivity. To stabilize world-space position, newly detected objects remain unanchored while their position is estimated; after repeated spatially consistent detections they transition to an anchored state and their transform is frozen, preventing continuous drift from noisy localization. Anchored objects are surfaced through lightweight indicators: white when interactable, pink on gaze fixation, green during manipulation (Fig. 4).

4.2.2 *Hand–Object Interaction.*

Hand tracking uses MRTK3 skeletal joint data at approximately 30 Hz. Grab and release are estimated from finger curl, hand posture, wrist proximity to anchored objects, and a short gaze-based disambiguation window for resolving nearby objects. During manipulation the object frequently occludes itself from the camera, making visual detection unreliable. Without additional handling, such tracking failures can interrupt action recognition and lead to incorrect procedural interpretations. The system therefore employs a *HandProxy* mechanism in which the manipulated object's transform follows the wrist joint rather than visual detections. By maintaining a continuous object representation throughout manipulation, the mechanism provides stable inputs for subsequent semantic action recognition and task-state tracking. Additional interaction features—wrist orientation, object tilt, pinch state, hand proximity, gaze fixation, and inter-object spatial relationships—are assembled continuously and streamed to the backend.

4.2.3 *Semantic Action Recognition.*

Because action recognition lies on the critical path of the interaction loop, the system must identify user actions with sufficiently low latency to support real-time procedural validation and adaptive guidance. While procedural actions can be recognized using end-to-end video recognition or vision–language models, such approaches introduce additional inference latency and typically require costly procedure-specific training data. Furthermore, their predictions are less directly interpretable for dependency-aware procedural validation. We therefore adopt an interaction-centric recognition strategy that exploits known object identities, spatial relationships, and temporal interaction constraints to infer semantic actions (*grab*, *place*, *pour*, *stir*, *press*) in real time. For example, *pour* requires sustained object tilt beyond approximately 45° toward a compatible target container, *grab* requires the wrist joint within roughly 10–15 cm of an anchored object, *stir* is detected from circular wrist trajectories, and *press* from an index-finger poke. Recognized actions are forwarded to the procedural state tracker for validation.

4.2.4 ***Dependency-Aware Validation and Outcome Classification.*** Each recognized action is matched against the valid transitions permitted by the current graph state. The tracker maintains active branches, completed steps, dependency-satisfaction state, and a history of invalid transitions. Because validation is performed relative to procedural dependencies rather than a fixed linear order, the same action can be interpreted differently depending on context—valid progress on one branch, but a premature attempt when its prerequisites are unmet.

Following validation, each action is assigned a procedural *outcome*, adapting the outcome-classification scheme of AMMA [23] to dependency-aware MR interaction. We distinguish five outcomes: *expected*, satisfying current dependencies; *premature*, attempted before prerequisites are complete; *repeat*, redundantly repeating a completed action; *harmless*, an off-protocol action with no procedural effect; *incorrect-recoverable*, a non-critical deviation; and *incorrect-unrecoverable*, a critical violation requiring corrective intervention. These outcomes provide a semantic interpretation of procedural events and enrich the procedural state maintained by the tracker. Together with task-progress information, dependency status, and interaction-history features, they are incorporated into the *Aggregated Manual Features* representation, which provides explicit procedural context for subsequent adaptive guidance generation.

4.3 Behavioral User-State Estimation

In parallel with semantic tracking, the system estimates *observable procedural difficulty* from temporal interaction behavior. **Observable task outcomes and user difficulty are not necessarily equivalent:** users may perform incorrect actions with little hesitation, while prolonged hesitation can occur even when actions are ultimately correct. The system therefore complements procedural outcomes with a behavioral estimate of user difficulty derived from multimodal interaction patterns. Rather than inferring latent cognitive load, emotion, or affect—constructs that are difficult to validate in unconstrained MR settings and often depend on intrusive sensing—procedural difficulty is operationalized as a behaviorally observable proxy for user struggle, consistent with behavioral confusion-detection approaches [18]. This signal distinguishes periods of fluent task execution from periods associated with struggle and serves as the implicit adaptation input that complements the explicit procedural context.

4.3.1 Data and Labeling. The estimator was trained on the HoloAssist dataset [9, 20], which provides synchronized egocentric gaze, hand-motion, and head-motion signals together with procedural interaction annotations across 25 task categories. Because no direct measurements of procedural difficulty are available in HoloAssist, labels were derived from procedural annotations that serve as observable indicators of user struggle: *struggling* windows from corrected and uncorrected wrong actions and instructor-correction events, and *stable-execution* windows from correct actions, with a temporal exclusion region around nearby errors to reduce ambiguous transitions. Instead of fixed sliding windows, we use event-centered segmentation, extracting a window of approximately ±2 s around each annotated event. Windows with insufficient valid frames or excessive tracking dropout were discarded, yielding a final dataset of 21,787 windows from 1,720 sessions.

4.3.2 Model and Generalization. We compared two representations on the same data. Aggregated statistical interaction features (e.g., hand-motion dynamics, pause ratios, gaze-motion entropy, head stability) were used with a gradient-boosted baseline (XGBoost), while raw multimodal temporal sequences (30 synchronized channels per window) were modeled with a bidirectional gated recurrent unit (BiGRU) [3]—two bidirectional GRU layers (64 and 32 units) followed by a dense projection with dropout. Under a matched train–test evaluation, the BiGRU substantially outperformed the engineered-feature baseline (macro F1 0.847 vs. 0.678), indicating that temporal behavioral dynamics carry information lost by aggregate summaries.

Task-identity features were intentionally excluded after preliminary analysis showed they encouraged task-specific memorization rather than generalizable behavioral modeling. To assess cross-task generalization—the deployment-relevant case, in which the target procedure is unseen during training—we additionally evaluated the BiGRU with Leave-One-Task-Out (LOTO) cross-validation across all 25 task categories, where it reached a macro F1 of 0.842 with a struggling-class recall of 0.95. The small gap from the train–test split (0.847) indicates that the model captured transferable behavioral dynamics rather than task-specific patterns.

4.3.3 Runtime Calibration and Inference. Because behavioral baselines vary substantially across individuals, calibration was introduced to reduce inter-user variability. Before adaptive assistance is enabled, a short per-session calibration phase estimates difficulty probabilities while generating no interventions, and a participant-specific baseline distribution is computed from these values. Intervention thresholds are initialized at approximately the 75th percentile of this distribution, and the runtime signal is smoothed with an exponential moving average to reduce rapid fluctuations. At runtime, the module continuously processes recent multimodal sequences and forwards the resulting difficulty estimate to the guidance layer as the implicit adaptation signal.

4.4 Adaptive Guidance Policy

The adaptive guidance module generates context-aware procedural assistance in two operational modes: *reactive*, where guidance is produced in response to user voice queries, and *proactive*, where the system autonomously initiates assistance when sustained user difficulty is detected. Both modes are driven by a runtime policy that fuses two independently computed inputs: a procedural context derived from the step tracker and a user-state estimate reflecting inferred task difficulty. The signals are computed in separate components and combined only at prompt construction time to form the final context for guidance generation.

4.4.1 *Language Model and Prompt Construction*.

Guidance is generated using a locally hosted instruction-tuned language model (Qwen2.5 7B Instruct) served through Ollama. Local deployment avoids cloud dependency during task execution and maintains inference latency within interactive bounds. The model conditions on both the aggregated procedural context and the estimated user-state signal, and produces a structured multimodal response consisting of natural-language instructions, synthesized speech output, and, when appropriate, a task-relevant object reference for visual highlighting. Response verbosity is modulated by the estimated user difficulty. When difficulty is elevated, the system generates more explicit instructions by elaborating the next step retrieved from the procedural step tracker, and may further simplify domain-specific terminology when needed to improve comprehensibility. Under stable execution conditions, responses remain concise and minimally intrusive. To avoid redundant assistance, the previous answer is supplied to the model with an instruction to take a different angle rather than repeat it. During development, three prompt configurations were compared offline: a protocol-only baseline, a version incorporating runtime procedural context, and a full configuration additionally including guidance history and the estimated user-state signal. Following the evaluation methodology of Qorbani et al. [15], guidance responses were assessed across diverse procedural scenarios using an LLM-as-a-judge protocol. The full configuration achieved the highest guidance-quality ratings and was therefore adopted in the deployed system.

4.4.2 *Guidance Delivery and Proactive Intervention*.

In addition to responding to explicit user requests, the system can proactively provide assistance when elevated procedural difficulty persists beyond a predefined duration threshold. To reduce unnecessary interruption, proactive intervention is gated through cooldown and interaction-state constraints, preventing repeated triggers during a single difficulty episode and suppressing intervention while the user is already engaged in conversation with the assistant. Critical procedural violations are handled through a separate low-latency path. When an action is classified as dangerous, the system immediately issues corrective feedback without invoking the language model. Successful step completions are similarly acknowledged through brief progress notifications announcing the next procedural step.

4.5 MR Guidance Interface

Guidance is delivered through coordinated verbal and visual channels within the MR environment. User speech is captured via keyword-triggered recognition on the headset, while system responses are synthesized using Azure Neural text-to-speech. Visual guidance reuses the runtime object-indicator system to provide spatially grounded feedback. Task-relevant objects referenced in guidance are highlighted using a pulsating yellow effect, enabling the system to direct the user's attention. A runtime guidance panel displays the current procedural step, overall progress, and contextual assistance text. Together, these modalities align natural-language instructions with spatial references, enabling the system to verbally identify objects while simultaneously indicating their physical location in the environment (Fig. 4).

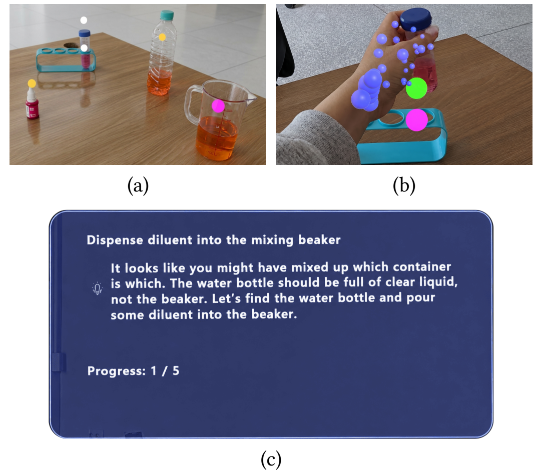

Figure 4: MR guidance interface components. (a) Workspace object indicators: ○ white = anchored and interactable, • pink = gazed at, • yellow = guidance highlight. (b) Hand–object interaction: • green = object grabbed, • blue = MRTK hand joints. (c) Adaptive guidance panel showing the current step, progress, and an LLM-generated instruction.

5 System Performance Evaluation

We evaluated the technical performance of the system: the reliability of its procedural interpretation, its runtime responsiveness, and the validity of the estimated difficulty signal during MR task execution. Metrics were computed over the recorded study sessions.

5.1 Semantic Tracking and Runtime Performance

We evaluated the tracking pipeline in terms of the procedural interpretation and its runtime responsiveness. *Semantic action recognition F1* measures how accurately low-level interactions are mapped to the correct semantic action; *outcome classification F1* measures how accurately each recognized action is assigned to its correct procedural outcome; and *step-completion accuracy* measures whether the tracker correctly advances through the task graph as steps are completed. Runtime responsiveness was characterized using three end-to-end measures: action-to-classification latency, deviation-to-feedback latency, and voice-query-to-response latency.

Table 1 summarizes the results. Semantic action recognition and outcome classification both achieved high accuracy (F1 0.97 and 0.96). Most importantly, step-completion accuracy reached 1.00, indicating that occasional action-level ambiguities did not propagate into incorrect task states and that procedural progress was tracked correctly throughout all evaluated sessions.

Table 1: Technical system performance across all evaluated sessions.

Metric	Mean ± SD
Semantic action recognition F1	0.97 ± 0.03
Outcome classification F1	0.96 ± 0.05
Step-completion accuracy	1.00 ± 0.00
Action-to-classification latency	58 ± 12 ms
Error-to-feedback latency	77 ± 65 ms
Voice query-to-response latency	4.8 ± 0.2 s

Runtime performance remained compatible with real-time interaction. Both the action-to-classification and deviation-to-feedback pipelines completed within tens of milliseconds, enabling near-real-time procedural monitoring and intervention. The full voice-query-to-response pipeline incurred higher latency due to speech processing and language-model inference (4.8 s mean), but remained compatible with the timescale of procedural task execution.

5.2 Validity of the Procedural Difficulty Signal

We further examined whether the runtime difficulty estimate produced by the user-state model (Sec. 4.3) is behaviorally meaningful—whether it rises before procedural deviations during actual task execution. For each participant, normalized difficulty was extracted from pre-action temporal windows preceding expected actions and procedural deviations. Difficulty was consistently higher before deviations than during stable execution (mean 0.063 vs. −0.002; $\Delta = 0.065$), a difference confirmed by a Wilcoxon signed-rank test ($W = 204$, $p < .001$) and a paired t-test ($t(19) = 4.58$, $p < .001$, $d_z = 1.02$, a large effect). These findings indicate that the estimated difficulty signal systematically increases prior to procedural deviations, supporting its use as an additional runtime cue for adaptive guidance. A per-participant visualization is provided in the supplementary material.

6 User Study

We conducted a controlled study to evaluate whether runtime adaptation improves procedural guidance relative to the same system in a non-adaptive mode, across tasks of differing complexity.

6.1 Participants

Twenty participants (graduate and undergraduate students with varying prior MR experience) were recruited from the university community, aged 20–35 (M = 23.7, SD = 3.4).

6.2 Study Design

The study used a counterbalanced mixed design with two system conditions and two task types. The **baseline** shared the same task graph, semantic tracking pipeline, instructional content, and MR interface as the adaptive system, and differed only by disabling the adaptive guidance behavior. In the baseline, guidance remained static: proactive intervention, user-state-driven adaptation, on-object highlighting, and spoken delivery—which together constitute the *output* of the adaptive policy rather than independent interface options—were inactive. We therefore evaluate the adaptive guidance condition as an **integrated package** against a static-guidance baseline on an otherwise identical tracking framework, and we interpret effects at the level of this package rather than attributing them to any single mechanism. The **simple** task involved familiar objects and straightforward instructions, whereas the **complex** chemistry-oriented task involved specialized laboratory equipment, domain-specific terminology, and a more complex dependency structure with greater potential for procedural errors.

Each participant experienced one baseline and one adaptive condition, and one simple and one complex task. Group A performed simple/baseline and complex/adaptive; Group B performed simple/adaptive and complex/baseline. Because no participant completed all four combinations, the design is a counterbalanced mixed design rather than a fully within-subject factorial design. This choice was deliberate: requiring each participant to perform the *same* task under both systems would introduce strong task-repetition learning and recall confounds that are difficult to remove in procedural tasks. The counterbalanced assignment instead keeps every task novel at first exposure.

6.3 Procedure

After eye-gaze calibration and a tutorial on hand-object interaction, voice interaction, and overall familiarization with the MR guidance interface, a short calibration phase established participant-specific baseline values for user-state estimation, collecting multimodal signals while generating no interventions. Participants then performed the assigned tasks while the system tracked procedural state and logged events, behavioral signals, and interventions. After each condition they completed NASA-TLX and a guidance-quality questionnaire, and provided qualitative feedback at the end.

7 Results and Analysis

We evaluated the impact of adaptive guidance on procedural task performance, recovery from deviations, and subjective user experience. Procedural error rate measures the proportion of actions resulting in procedural deviations, while struggling rate measures the proportion of runtime intervals classified as elevated procedural difficulty by the user-state estimator. Recovery latency captures the time required to return to a valid task state following a deviation. Subjective outcomes were assessed using NASA-TLX and a post-task guidance-quality questionnaire.

Behavioral and subjective measures were analyzed using linear mixed-effects models (LMMs) with System (Baseline vs. Adaptive), Task (Simple vs. Chemistry), and their interaction as fixed effects and participant as a random intercept.

7.1 Behavioral Impact of Adaptive Guidance

Both behavioral measures revealed a significant main effect of task complexity (error rate: $\beta = 0.361$, $p < .001$; struggling rate: $\beta = 0.634$, $p < .001$), indicating that the complex task imposed substantially greater procedural demands than the simple task.

More importantly, a significant System × Task interaction was observed for both measures (error rate: $\beta = -0.287$, $p < .001$; struggling rate: $\beta = -0.232$, $p < .001$), indicating that the effect of

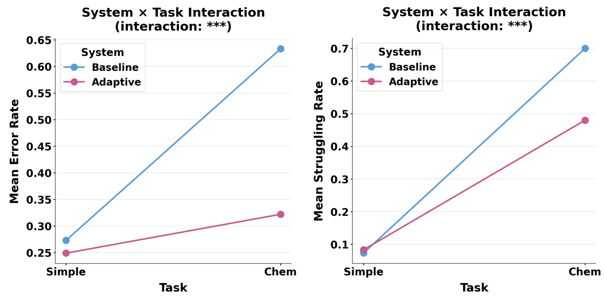

Figure 5: Effect of adaptive guidance on error rate (left) and struggling rate (right) across task conditions. Both measures increased sharply under baseline during the complex task, while adaptive guidance showed a substantially smaller increase ($p < .001$).

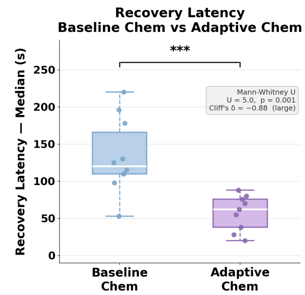

Figure 6: Recovery latency following procedural deviations during the chemistry-oriented task. Adaptive guidance substantially reduced time to recovery ($U = 5$, $p = .001$, Cliff's $\delta = -0.875$).

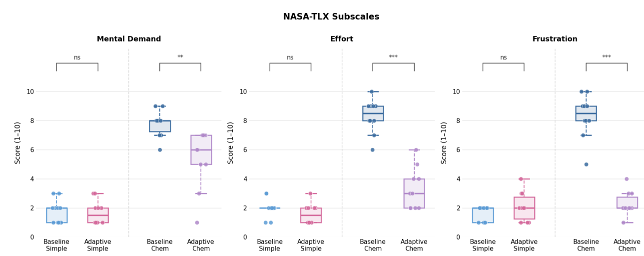

Figure 7: NASA-TLX subscale scores across conditions. Effort and frustration showed the largest reductions under adaptive guidance during the complex task ($p < .001$).

adaptive guidance depended on task complexity. As shown in Fig. 5, error rates and struggling rates increased substantially under the baseline condition when moving from the simple to the complex task, whereas both measures remained comparatively lower under adaptive guidance.

Follow-up analyses showed no significant differences between adaptive and baseline guidance in the simple task (error rate: $\beta = -0.025$, $p = .619$; struggling rate: $\beta = 0.009$, $p = .848$). In contrast, adaptive guidance reduced both procedural deviations and periods of elevated difficulty during the complex task. Together, these findings suggest that the benefits of adaptation emerge primarily when procedures involve greater complexity and a larger space of potential deviations.

7.2 Recovery After Procedural Deviations

To assess whether adaptive guidance facilitated recovery from procedural deviations, we measured recovery latency, defined as the elapsed time between the occurrence of a procedural deviation and the completion of the next valid task step.

Because no significant behavioral differences were observed during the simple task, the analysis focused on the complex task condition. As shown in Fig. 6, participants recovered substantially faster under adaptive guidance (mean 56.46 s) than under baseline guidance (mean 132.22 s). A Mann–Whitney U test confirmed a significant difference ($U = 5$, $p = .0009$), with a large effect size (Cliff's $\delta = -0.875$). These results indicate that adaptive guidance not only reduced procedural difficulty, but also helped participants return to the intended workflow more quickly after deviations occurred.

A post-hoc analysis of adaptive sessions further showed that 74.6% of proactive interventions were immediately followed by successful return to the intended workflow, suggesting that generated guidance was contextually appropriate and actionable in the majority of cases.

7.3 Perceived Workload and Guidance Quality

Subjective workload (NASA-TLX Core) and perceived guidance quality (GQI: mean of clarity, confidence, helpfulness, and explanation understandability) were analyzed using the same LMM structure. Both measures showed a significant System × Task interaction (NASA-TLX: $\beta = -4.467$, $p < .001$; GQI: $\beta = +4.267$, $p < .001$). Under the baseline condition, the complex task produced a sharp workload increase and a corresponding drop in guidance quality; the adaptive condition showed substantially reduced workload and consistently high guidance quality during the complex task, with no significant differences during the simple task (NASA-TLX: $\beta = 0.000$, $p = 1.000$; GQI: $\beta = -0.150$, $p = .580$). Subscale analysis indicated that the largest reductions occurred in effort and frustration (Fig. 7).

7.4 Qualitative Observations

Qualitative feedback generally aligned with the quantitative findings. Participants reported that adaptive guidance was particularly helpful during the chemistry task, where unfamiliar terminology and increased procedural complexity produced higher perceived difficulty. Visual object highlighting was frequently described as the most effective guidance modality, helping participants associate procedural instructions with physical laboratory objects.

Representative runtime interactions illustrate the system's ability to handle imperfect input and adapt to execution context. When a participant queried an unfamiliar term that was incorrectly transcribed as "dilutant," the system correctly inferred the intended referent and responded: *"Diluent, in this case, is just water. We use it to mix with the solution,"* while simultaneously highlighting the relevant objects in the MR scene. When another query contained the transcription error "bigger" instead of "beaker," the system inferred the correct referent and issued a safety-oriented response: *"No, you should not drink the liquid in the beaker. It is a colored solution used for the experiment and may contain chemicals."* During a prolonged struggling episode, rather than repeating the protocol description, the system decomposed the step into sequential actions: *"First, pick up the water bottle. Hold the beaker steady with your other hand and slowly tilt the bottle."*

Participants expressed mixed preferences regarding guidance modality. Several preferred reading instructions directly from the guidance panel, citing the ability to scan and revisit text at their own pace. Short voice-based notifications—step-completion confirmations and safety alerts—were nonetheless valued during active manipulation, when visual attention was directed at physical objects.

8 Discussion

8.1 Implications for Adaptive MR Guidance

The separation between explicit procedural outcomes and implicit difficulty estimation proved important in practice, as the two signals can diverge during interaction. Users may execute incorrect actions with high confidence, or hesitate while ultimately completing a step correctly. Treating both as complementary inputs allowed the system to remain responsive under these mismatched conditions, rather than relying on either procedural correctness or behavioral difficulty alone.

The complexity-dependent effects observed in the study suggest that the benefits of adaptation become increasingly important as procedural and semantic complexity grows. While static guidance may be sufficient for relatively simple procedures, adaptive guidance appears particularly valuable in tasks involving larger action spaces, domain-specific concepts, and greater opportunities for procedural deviation.

More broadly, the results suggest that adaptive MR assistance benefits from combining knowledge of what the user is doing with evidence of how they are interacting. Procedural state alone provides an incomplete view of user needs, while behavioral difficulty signals lack task context; integrating both may offer a practical foundation for future closed-loop guidance systems.

8.2 Limitations

The evaluation involved 20 participants performing two task types, and component-level ablations were not conducted; the results should therefore be interpreted as system-level evidence for the integrated adaptive framework rather than as causal evidence for any individual mechanism. We emphasize that adaptive content, proactive timing, and multimodal delivery are deliberately coupled in our design: adaptive guidance has little meaning without a channel to deliver it. The present study therefore targets system-level efficacy of this integrated package, which is the deployment-relevant question for a first end-to-end evaluation; disentangling the relative contribution of each mechanism is a goal for component-level ablation. Importantly, the benefit of adaptive guidance appears *consistently and in the same direction* across five independent measures (error rate, struggling rate, recovery latency, workload, and guidance quality), a convergent pattern that is difficult to explain by any single delivery channel alone.

The difficulty estimator was pretrained on HoloAssist data and deployed with participant-specific calibration; while the validation results indicate that the signal is behaviorally meaningful during task execution, further evaluation on target-domain data would strengthen its generalizability. Additionally, because each participant experienced one baseline and one adaptive condition, results should be interpreted as evidence from a counterbalanced mixed design rather than a full repeated-measures factorial experiment; the LMM accounts for participant-level variability through random intercepts, but does not fully compensate for the incomplete factorial exposure—though the consistency of effects across measures makes a purely group-driven explanation unlikely. Finally, guidance quality remains bounded by the capabilities of the underlying language model. Although local deployment enabled interactive operation without cloud connectivity, occasional oversimplification and reduced explanatory depth were observed for more complex user queries.

8.3 Future Work

Future work should investigate component-level ablations to isolate the effects of individual adaptation mechanisms, and evaluate the framework across broader procedural domains and longer task sequences. Stronger multimodal grounding and domain-specific fine-tuning of the language model may further improve guidance reliability. An additional direction is adaptive selection of guidance modality and presentation style. Participant feedback suggested differing preferences between textual and voice-based guidance, indicating that future systems may benefit from adapting not only guidance content, but also how assistance is delivered. Extending the framework to multi-user or instructor-in-the-loop scenarios represents a further direction for deployment-oriented research.

9 Conclusion

We presented a real-time multimodal adaptive MR system for procedural task guidance that integrates semantic task-state tracking, behavioral user-state estimation, and LLM-based guidance generation within a closed-loop architecture on HoloLens 2. A controlled user study with 20 participants demonstrated that adaptive guidance substantially reduced procedural errors, periods of elevated difficulty, and recovery time during complex task execution. Together, these results indicate that combining explicit procedural context with implicit behavioral user-state information enables more effective adaptive assistance than task-state awareness alone, and represents a promising direction for future intelligent MR guidance systems.

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